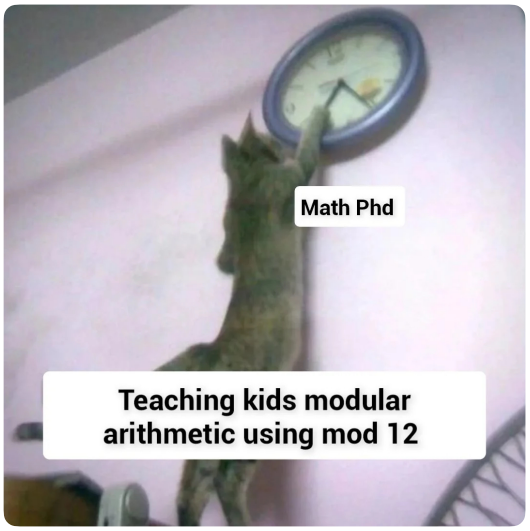
Week 7

Number Theory:

- Divisions
- Primes
- Modular Arithmetic
- Diffie - Hellman



Kahoot

Bonus Week 6

Solution: Two sets A and B are *equinumerous*, denoted $A \sim B$, if there exists a bijection $A \to B$.

2. (2 points) ($\star$)

Prove that the union $S \stackrel{\text{def}}{=} \cup_{i \in \mathbb{N}} A_i$ of a (countably) infinite sequence $A_0, A_1, A_2, \ldots$ of countable sets is countable.

Solution: We know that a set A is countable if and only if it is finite or if $A \sim \mathbb{N}$. In both cases we can do a (possibly infinitely long) enumeration of all elements of A. If A is finite, then we can trivially enumerate all elements in a finite number of steps. If $A \sim \mathbb{N}$, then also $\mathbb{N} \sim A$, as $\sim$ is symmetric (cf. Lemma 3.15). Thus, there exists a bijection $f : \mathbb{N} \to A$. We can hence enumerate all elements of A as $f(1), f(2), f(3), \ldots$.

We can now define an injection $g : S \to \mathbb{N} \times \mathbb{N}$ as follows: We write down all elements of $S = \cup_{i \in \mathbb{N}} A_i$ in a matrix. The entry (i, j) of this matrix is the j-th element of the enumeration of elements of set A_i. We can then define g to map each element $s \in S$ to its index (i, j) in this matrix. Note that maybe, for some sets A_i, the enumeration of its elements is finite and that the same element might appear in different sets A_i, A_j, possibly leading to duplicates in our matrix. If the enumeration of the elements of some set A_i has ended or if an element already appeared in an earlier row in the matrix, we can simply leave the corresponding entry of the matrix empty. Thus, we can ensure that g is a well-defined function. In this mapping, no two elements get mapped to the same tuple $(i, j) \in \mathbb{N} \times \mathbb{N}$, as by construction no two elements get written down at the same entry (i, j) of the matrix (i.e. the mapping is injective). It hence holds that $S \preceq \mathbb{N} \times \mathbb{N}$ and as $\mathbb{N} \times \mathbb{N} \preceq \mathbb{N}$, we know that $S \preceq \mathbb{N}$ (cf. Lemma 3.15, Theorem 3.19). Hence, by Theorem 3.17, S is countable.

$S \to \mathbb{N} \times \mathbb{N}$
$s \mapsto (i, j)$
(s is the jth elem. of A_i)

Why can you assign a j to every elem.

Alternatively:
$g: s \mapsto (i, f_i(s))$
f_i is the injection from A_i to $\mathbb{N}$

Then prove g inj.
$(i, f_i(s)) = (j, f_j(s'))$
$i = j \Rightarrow f_i = f_j$
$f_i(s) = f_i(s')$
$s = s'$

3. (3 points) ($\star\star$)

Consider the set

$$S \stackrel{\text{def}}{=} \left\{ f \in \mathbb{N}^{\mathbb{N}} \quad | \quad \forall x \forall y \quad \Big(x \leq y \to \big(f(x) \leq f(y) \big) \Big) \right\}.$$

Prove or disprove that the set S is countable.

Solution: We prove that S is uncountable by exhibiting an injection from $\{0,1\}^\infty$ to S. For any $b \in \{0,1\}^\infty$, let b_i for $i \in \mathbb{N}$ denote the i-th bit of b. We define the function $f_b : \mathbb{N} \to \mathbb{N}$ by

$$f_b(x) \stackrel{\text{def}}{=} \sum_{i=0}^{x} b_i.$$

Pretty good.

We define the function $g : \{0,1\}^\infty \to S$ by $g(b) \stackrel{\text{def}}{=} f_b$. It is easy to verify that $f_b \in S$ for any $b \in \{0,1\}^\infty$. For any $x \in \mathbb{N}, y \in \mathbb{N}$ with $x \leq y$ we have

don't forget this

$$f_b(y) - f_b(x) = \sum_{i=0}^{y} b_i - \sum_{i=0}^{x} b_i = \sum_{i=x+1}^{y} b_i \geq 0$$

and hence

$$f_b(y) \geq f_b(x).$$

We show that g is injective: Let $b, b' \in \{0,1\}^\infty$ be arbitrary and assume $b \neq b'$. This

Division

Theorem 4.1 (Euclid). *For all integers a and $d \neq 0$ there exist unique integers q and r satisfying*

$$a = dq + r \quad \text{and} \quad 0 \leq r < |d|.$$

Definition 4.2. For integers a and b (not both 0), an integer d is called a *greatest common divisor*[6] of a and b if d divides both a and b and if every common divisor of a and b divides d, i.e., if

$$d \mid a \ \wedge \ d \mid b \ \wedge \ \forall c \big((c \mid a \ \wedge \ c \mid b) \ \rightarrow \ c \mid d\big).$$

a = 4 b = 6
d = ±2

Definition 4.3. For $a, b \in \mathbb{Z}$ (not both 0) one denotes the unique positive greatest common divisor by $\gcd(a, b)$ and usually calls it *the* greatest common divisor. If $\gcd(a, b) = 1$, then a and b are called *relatively prime*[7].

gcd(4,6) = 2

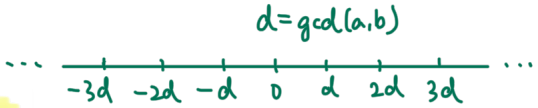
Lemma 4.2. *For any integers m, n and q, we have*

$$\gcd(m, R_m(n)) = \gcd(m, n),$$

$$\gcd(m, n - qm) = \gcd(m, n).$$

Definition 4.4. For $a, b \in \mathbb{Z}$, the *ideal generated by a and b*[8], denoted (a, b), is the set

$$(a, b) \stackrel{\text{def}}{=} \{ua + vb \mid u, v \in \mathbb{Z}\}.$$



Lemma 4.3. *For $a, b \in \mathbb{Z}$ there exists $d \in \mathbb{Z}$ such that $(a, b) = (d)$.*

Lemma 4.4. *Let $a, b \in \mathbb{Z}$ (not both 0). If $(a, b) = (d)$, then d is a greatest common divisor of a and b.*

Corollary 4.5. *For $a, b \in \mathbb{Z}$ (not both 0), there exist $u, v \in \mathbb{Z}$ such that*

$$d \rightsquigarrow \gcd(a, b) = ua + vb.$$

Definition 4.5. The *least common multiple l* of two positive integers a and b, denoted $l = \operatorname{lcm}(a, b)$, is the common multiple of a and b which divides every common multiple of a and b, i.e.,

$$a \mid l \ \wedge \ b \mid l \ \wedge \ \forall m\big((a \mid m \ \wedge \ b \mid m) \ \rightarrow \ l \mid m\big).$$

HS 19

c) $(\star\,\star)$ Prove that for all $a, b, c \in \mathbb{Z} \setminus \{0\}$, if $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$. *(6 Points)*

More intuitive but error prone:

Case 1: $a = 1 \Rightarrow a \mid c$

Case 2: $a \neq 1$

All factors of a are not in b.

But $a \mid bc$ so those factors are all in c.

$\Rightarrow a = \pm \prod_i p_i^{e_i}$

$b = \pm \prod_i p_i^{f_i}$

$c = \pm \prod_i p_i^{g_i}$

$bc = \pm \prod_i p_i^{f_i + g_i}$

$\forall i: \ e_i \neq 0 \Rightarrow f_i = 0 \Rightarrow f_i + g_i = g_i$

need more steps to prove

$a \mid bc \qquad g_i \geq e_i \qquad a \mid c$

Better proof (also shorter)

$c = c \cdot 1$

$= c \cdot \gcd(a, b)$

$= c\,(ua + vb) \quad u, v \in \mathbb{Z}$

$= cua + vbc$

$= cua + vka \quad k \in \mathbb{Z}$

$= (cu + vk)\,a$

$\Rightarrow a \mid c$

$(cu + vk \in \mathbb{Z})$

$c = ka$

$c \in (a)$

$c \in (\gcd(a, b))$

1

$c = c \cdot 1$

$(a \mid bc)$

Euclidean Algorithm & Extended Euclidean Algorithm

Good to know, but not relevant for the exam.

Lemma 4.2: $\gcd(553, 142)$

$$553 = 3 \cdot 142 + 127 \quad \gcd(142, 127)$$
$$142 = 1 \cdot 127 + 15 \quad \gcd(127, 15)$$
$$127 = 8 \cdot 15 + 7 \quad \gcd(15, 7)$$
$$15 = 2 \cdot 7 + 1 \quad \gcd(7, 1)$$
$$7 = 7 \cdot 1 + 0 \quad 1$$

Euclidean Algo

$15 - 2(127 - 8 \cdot 15) = 17 \cdot 15 - 2 \cdot 127$

$15 - 2 \cdot 7$

1

$74 \cdot 142 - 19 \cdot 553$

Extended Euclidean Algo

Prime Numbers

Definition 4.6. A positive integer $p > 1$ is called *prime* if the only positive divisors of p are 1 and p. An integer greater than 1 that is not a prime is called *composite*[9].[10]

Theorem 4.6. *Every positive integer can be written uniquely (up to the order in which factors are listed) as the product of primes.*[11]

$$a = \prod_i p_i^{e_i} \quad \text{and} \quad b = \prod_i p_i^{f_i}.$$

$$\gcd(a,b) = \prod_i p_i^{\min(e_i, f_i)}$$

$$\operatorname{lcm}(a,b) = \prod_i p_i^{\max(e_i, f_i)}.$$

$a = 24 \quad b = 180$

$a = 2^3 \cdot 3^1 \cdot 5^0$

$b = 2^2 \cdot 3^2 \cdot 5^1$

$\Rightarrow \gcd(a,b) = 2^2 \cdot 3^1 = 12$

$\operatorname{lcm}(a,b) = 2^3 \cdot 3^2 \cdot 5^1 = 360$

It is easy to see that

$$\gcd(a,b) \cdot \operatorname{lcm}(a,b) = ab$$

because for all i we have

$$\min(e_i, f_i) + \max(e_i, f_i) = e_i + f_i.$$

Modular Arithmetic

Definition 4.8. For $a, b, m \in \mathbb{Z}$ with $m \geq 1$, we say that a *is congruent to* b *modulo* m if m divides $a - b$. We write $a \equiv b \pmod{m}$ or simply $a \equiv_m b$, i.e.,

$$a \equiv_m b \overset{\text{def}}{\iff} m \mid (a - b).$$

Lemma 4.13. *For any $m \geq 1$, $\equiv_m$ is an equivalence relation on $\mathbb{Z}$.*

Lemma 4.14. *If $a \equiv_m b$ and $c \equiv_m d$, then*

$$a + c \equiv_m b + d \quad \text{and} \quad ac \equiv_m bd.$$

$$a = b \implies a \equiv_m b$$

$$a \not\equiv_m b \implies a \neq b.$$

Lemma 4.16. *For any $a, b, m \in \mathbb{Z}$ with $m \geq 1$,*

(i) $a \equiv_m R_m(a)$.

(ii) $a \equiv_m b \iff R_m(a) = R_m(b)$.

Corollary 4.17. *Let $f(x_1, \ldots, x_k)$ be a multi-variate polynomial in k variables with integer coefficients, and let $m \geq 1$. Then*

$$R_m\big(f(a_1, \ldots, a_k)\big) = R_m\big(f(R_m(a_1), \ldots, R_m(a_k))\big).$$

$\mathbb{Z}_m = \{0, 1, \cdots, m-1\}$

Compute $R_{18}(37^{42})$

$= R_{18}(R_{18}(37)^{42}) = R_{18}(1^{42}) = 1$

Compute $R_8(3^{2017})$ $\quad 3 \cdot 3^{2016}$

$= R_8(3 \cdot 9^{1008}) = R_8(3 \cdot R_8(9)^{1008}) = R_8(3 \cdot 1^{1008}) = 3$

Compute $R_{18}(17^5)$ $\quad 17 \equiv_{18} -1$

$= R_{18}((-1)^5) = R_{18}(-1) = 17$

Show that 10 divides $(43^{43} - 17^{17})$

$R_{10}(43^{43}) = R_{10}(3^{43}) = R_{10}(3 \cdot 9^{21}) = R_{10}(3 \cdot (-1)^{21})$

$R_{10}(17^{17}) = R_{10}(7^{17}) = R_{10}(\underbrace{(-1)^8}_{9} \cdot 7) = R_{10}(7) = 7$

$R_{10}(43^{43} - 17^{17}) = R_{10}(7 - 7) = 0 \Rightarrow 10 \mid 43^{43} - 17^{17}$

Show that equation $x^3 - x = y^2 + 1$ has no solution for $x, y \in \mathbb{Z}$.

$R_2(x)$	$R_2(x^3-x)$
0	0
1	0

$R_2(y)$	$R_2(y^2+1)$
0	1
1	0

$R_3(x)$	$R_3(x^3-x)$
0	0
1	0
2	0

$R_3(y)$	$R_3(y^2+1)$
0	1
1	2
2	2

Try

$m = 2$

$m = 3$

$m = 4$

$m = 5$

$\vdots$

$\forall x, y \in \mathbb{Z} \quad R_3(x^3 - x) \neq R_3(y^2 + 1)$

$\Rightarrow \quad x^3 - x \neq y^2 + 1$

$\Rightarrow \quad x^3 - x = y^2 + 1$ has no solution.

Multiplicative Inverse

Lemma 4.18. *The congruence equation*

$$ax \equiv_m 1$$

has a solution $x \in \mathbb{Z}_m$ if and only if $\gcd(a, m) = 1$. The solution is unique.

$\gcd(a, m) = 1 = ua + vm \qquad u, v \in \mathbb{Z}$

$\equiv_m ua$

$u = a^{-1}$

Compute the inverse of 142 modulo 553

Euclidean Algorithm & Extended Euclidean Algorithm

$1 = (74) \cdot 142 - 19 \cdot 553$

Lemma 4.2: $\gcd(553, 142)$

$74 \cdot 142 - 19 \cdot 553$

$\vdots$

$15 - 2(127 - 8 \cdot 15) = 17 \cdot 15 - 2 \cdot 127$

$15 - 2 \cdot 7$

1

Extended Euclidean Algo

$$\begin{aligned} 553 &= 3 \cdot 142 + 127 \\ 142 &= 1 \cdot 127 + 15 \\ 127 &= 8 \cdot 15 + 7 \\ 15 &= 2 \cdot 7 + 1 \\ 7 &= 7 \cdot 1 + 0 \end{aligned}$$

$\gcd(553, 142)$

$\gcd(142, 127)$

$\gcd(127, 15)$

$\gcd(15, 7)$

$\gcd(7, 1)$

1

Euclidean Algo

Chinese Remainder Theorem

Theorem 4.19. *Let $m_1, m_2, \ldots, m_r$ be pairwise relatively prime integers and let $M = \prod_{i=1}^{r} m_i$. For every list $a_1, \ldots, a_r$ with $0 \le a_i < m_i$ for $1 \le i \le r$, the system of congruence equations*

$$\begin{aligned} x &\equiv_{m_1} a_1 \\ x &\equiv_{m_2} a_2 \\ &\cdots \\ x &\equiv_{m_r} a_r \end{aligned}$$

for x has a unique solution x satisfying $0 \le x < M$.

~~*Proof.* Let $M_i = M/m_i$. Hence $\gcd(M_i, m_i) = 1$ because every factor m_k (where $k \neq i$) of M_i is relatively prime to m_i, and thus so is M_i. Thus~~ there exists an N_i satisfying

$$M_i N_i \equiv_{m_i} 1.$$

~~Note that for all $k \neq i$ we have $M_i \equiv_{m_k} 0$ and thus~~

~~$M_i N_i \equiv_{m_k} 0.$~~

~~Therefore~~

$$\sum_{i=1}^{r} a_i M_i N_i \equiv_{m_k} a_k$$

~~for all k.~~ Hence the integer x defined by

$$x = R_M \left(\sum_{i=1}^{r} a_i M_i N_i \right)$$

Method →

HS 17

b) ($\star\,\star$) Compute $R_{63}(100^{1207})$. $= x$ *(3 Points)*

Hint: Compute first $R_7(100^{1207})$ and $R_9(100^{1207})$.

$x \equiv_{63} 100^{1207}$

$x \equiv_7 100^{1207} \equiv_7 2$

$x \equiv_9 100^{1207} \equiv_9 1$

$$R_7(100^{1207}) = R_7(R_7(100)^{1207}) = R_7(2^{1207})$$
$$= R_7((2^3)^{400} \cdot 2^7) = R_7(R_7(2^3)^{400} \cdot R_7(2^7)) = R_7(1 \cdot R_7(2^7))$$
$$= R_7(R_7(2^3)^2 \cdot 2) = R_7(1 \cdot 2) = 2$$

$$R_9(100^{1207}) = R_9(R_9(100)^{1207}) = R_9(1^{1207}) = 1$$

$x \equiv_7 2 \qquad m_1 = 7 \qquad a_1 = 2$

$x \equiv_9 1 \qquad m_2 = 9 \qquad a_2 = 1$

$M = 63$

Therefore

$$\sum_{i=1}^{r} a_i M_i N_i \equiv_{m_k} a_k$$

for all k. Hence the integer x defined by

$$x = R_M\left(\sum_{i=1}^{r} a_i M_i N_i\right)$$

$M_1 = M / m_1 = 9$

$M_2 = 7$

$M_1 N_1 \equiv_7 1 \Rightarrow N_1 = 4$

$M_2 N_2 \equiv_9 1 \Rightarrow N_2 = 4$

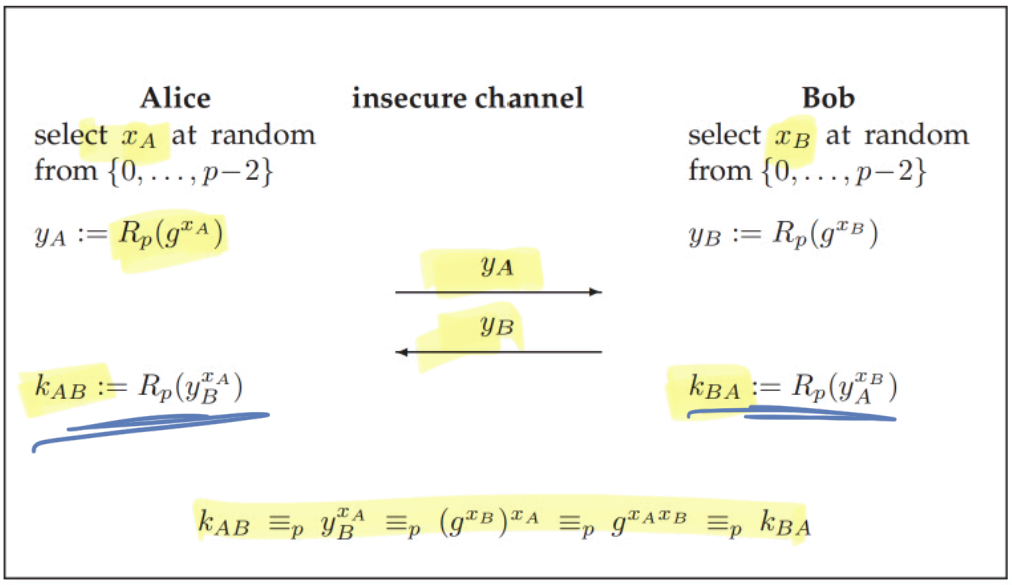
$x = R_{63}(2 \cdot 9 \cdot 4 + 1 \cdot 7 \cdot 4)$

$= R_{63}(72 + 28)$

$= R_{63}(100)$

$= 37$

Diffie–Hellman key



Actually works even when p is not prime.

HS 20

$y_A, y_B \in \{1, 2, 4, 8\}$

c) (⋆) Alice and Bob execute the Diffie-Hellman protocol using the subgroup of $\langle \mathbb{Z}_{15}^*; \odot \rangle$ that is generated by $g = 2$. Alice sends the message $y_A = 8$. They agree on the secret key $k_{AB} = 2$. *(3 Points)*

What message did Bob send? y_B **8**

$2 = k_{AB} = k_{BA} = R_{15}(8^{x_B})$ $\quad x_B = \not{x}\ \not{8}\ 3$

$8^3 = 64 \cdot 8$

$\equiv_{15} 4 \cdot 8$

$\equiv_{15} 2$

$y_B = R_{15}(2^3) = 8$